Name: P. Poorna Chandra
Regno: 192472287
Course code: ITA0502

(Assessment Tool -3)

Slot-C Co2 AT-3 Class Test -1

1. Image Enhancement in the Spatial Domain.

20/25

Image Enhancement in the process of improving the quality of an image so that it becomes clearer and easier to analyze. In the spatial domain, enhancement is done by directly changing the pixel values of the image.

* Gray-Level Transformation: improve brightness and contrast (Image Negative, contrast stretching, Gamma Correction)
* Histogram processing: improves image contrast using Histogram Equalization.
* Spatial filtering: uses filters to remove noise and enhancement edges.
* Smoothing filters reduce noise.
* Sharpening filters improve edge clarity.

Applications:-
* Medical Imaging.
* Satellite Images.
* CCTV Surveillance
* Industrial inspection.

Conclusion:- Spatial domain techniques improve image quality by directly modifying pixel values.

2. Differentiate between Image Smoothing and Image Sharpening.

Image Smoothing and image Sharpening are two important image enhancement techniques. Smoothing removes noise, while Sharpening highlights important details and edges.

Image Smoothing.

* Removes unwanted noise.
* produces a Smooth image
* Uses Mean, Gaussian, and Median filters.
* Applied before segmentation and object detection.

Image Sharpening:-

* Enhances edges and fine details.
* produces a clearer image.
* Uses Laplacian, sobel, and prewitt filters.
* Applied in fingerprint recognition, OCR, and medical imaging

Difference:-

Image Smoothing	Image Sharpening.
Removes noise	Enhances edges.
Blurs image	Sharpens image.
Uses Low-pass filters	Uses High-pass filters.

Application.

* Smoothing : Noise removal.
* Sharpening : Edge enhancement.

Conclusion:-

Smoothing improves image quality by reducing noise, while Sharpening improves image clarity by enhancing edges.

3. Image Enhancement in the Frequency Domain.

In the frequency domain an image is converted into frequency components using the fourier Transform. Filters are applied to improve the image, and then the image is conver-ted back.

1. Fourier Tramform:
 It converts image from spatial domain to frequency domain

2. Low-pass filter.
 => Removes high frequency noise.
 => produces a smooth image.

3. Higher-pass filter.
 => Enhances edges and fine details,
 => produces a sharper image.

4. Comparison
 => LPF reduce noise.
 => HPF increases sharpness.

5. Applications.
 Medical imaging.
 Statellite imaging.
 Face recognition
 Industrial inspection.

~~Conclusion~~:

Frequency domain enhancement improves image quality using LPF for smoothing and HPF for sharpening.

4. Image Segmentation.

Image Segmentation is the process of dividing an image into different regions or objects. It helps identify and analyze objects separately from the background.

1. Point Detection. :-

* It Detects individual pixels.

2. Line Detection. :-

=> Detects straight lines.

3. Edge Detection. :-

=> It Detects object boundaries.

4. Thresholding. :-

* It separates objects based on intensity values

5. Region - Based Segmentation

=> Groups neighboring pixels with similar properties.

Advantages:-

* Easy Object detection.
* Better image analysis.
* improved accuracy.

Applications

=> Medical imaging
=> Facial recognition
=> Traffic monitoring
=> Industrial inspection

Conclusion:-

Edge detection and region - based segmentation provide accurate object detection and are widely used in real-world applications.

5. Edge Detection, Edge Linking, and Boundary Detection

Edge detection, edge linking, and Boundary detection are important techniques used to identify object shapes and boundaries. They improve object recognition and defect detection.

1. Edge Detection :- Detects sudden intensity changes.
   Methods : sobel, prewitt, canny.

2. Edge Linking :- Connects broken edge segmentation produces continuous object boundaries.

3. Boundary Detection :- Identifies complete object outlines.
   Helps measure object size and shape.

4. Importance
   → Improves object recognition.
   → Detects defects
   → used in medical diagnosis.
   → used in quality inspection.

Ex:- In a factory, edge detection finds cracks, edge linking joins broken crack edges, and boundary detection measures the crack size.

Conclusion:-

Edge detection identifies edges, edge linking connects them, and boundary detection forms complete boundaries, making object recognition and defeat detection more accurate.